
AMBIENT
SYSTEMS
Reimagining the future of smart cities

40%

of global CO2 emissions is attributed to buildings

Climate Impact will be largely felt by the real estate asset managers

Climate Risk can reduce asset value by 20% and annual returns by 40%

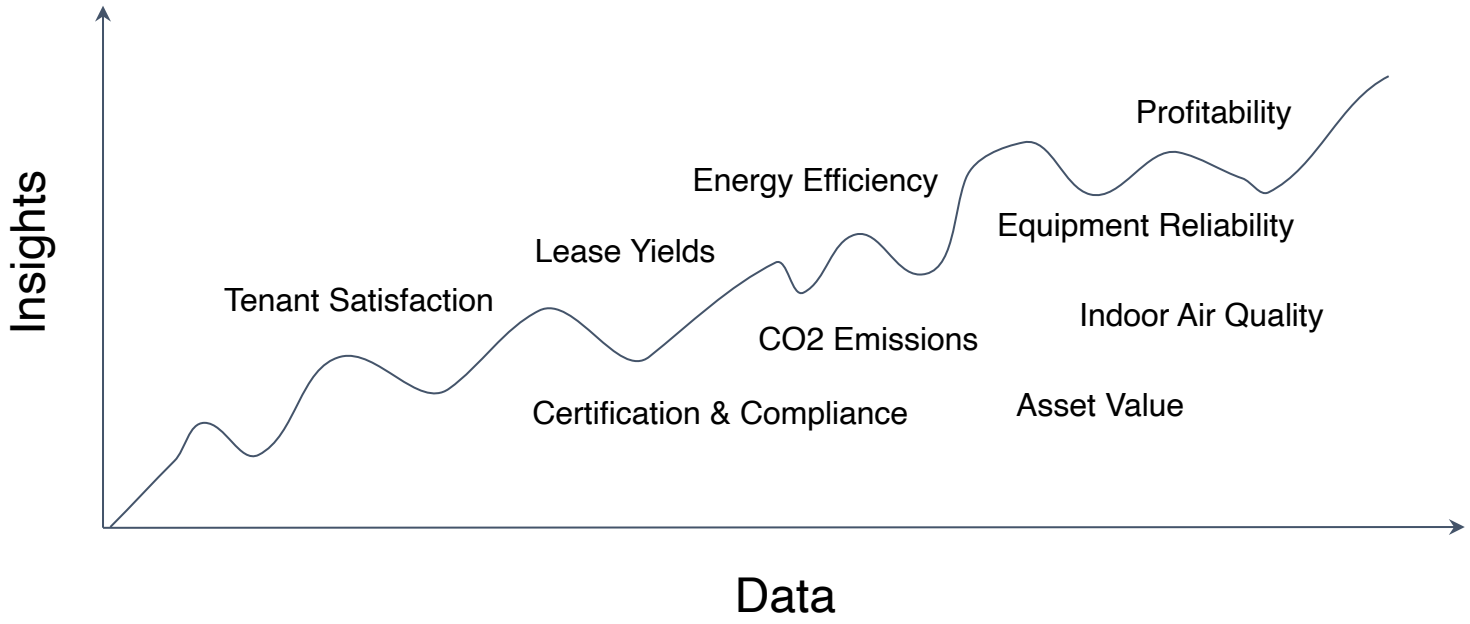
Risk to Asset Owner

- Insurance Premiums for ’Risky’ Properties
- Higher Facility Management Costs
- Risk of Default by Carbon-Intensive Occupier
- Cost of refurbishing inefficient facilities
- Repairing physical damage (e.g., floods, storms)
- Increased local taxes for community protection
- Impact on the local economy

Risk to Occupier

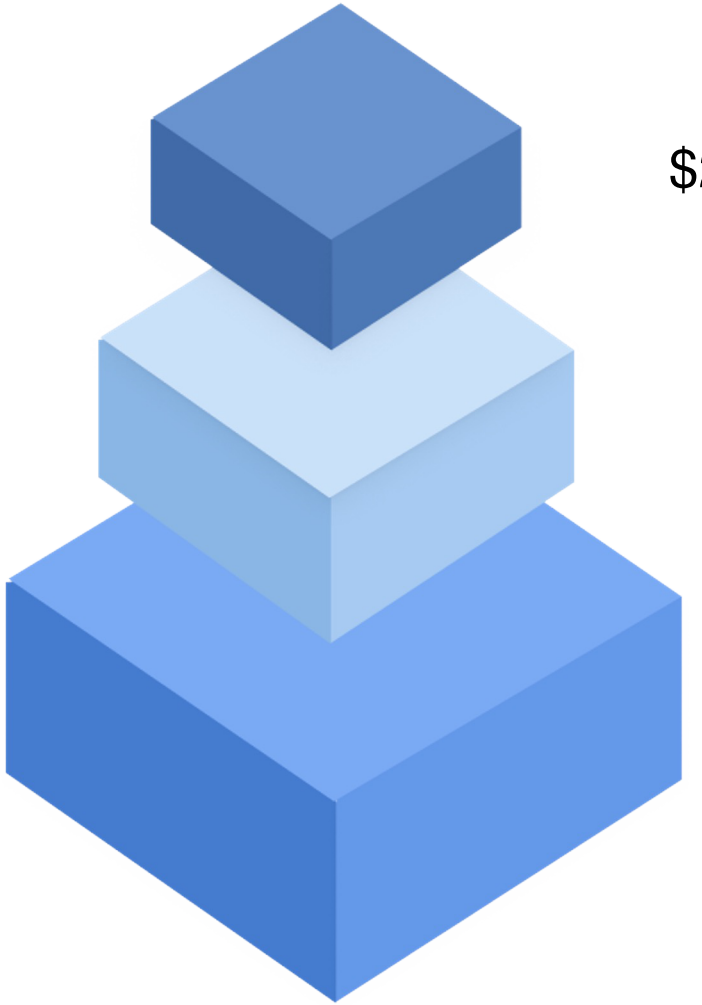
- Carbon Taxes
- Increased Utility Costs
- Risk of defaulting on lease due to increased fees

Data Science: empowering real estate asset managers to overcome climate risk



Identify Overpriced or Underpriced Assets
+
Operationalize Your Climate Resilience
+
Attract ‘Green’ Capital and Tenants

Minimizing Climate Risk impact on portfolio valuation and annual returns



$2.6B

Serviceable Obtainable Market (SOM)

Ambient's Potential Market Share

- Top 10 Asset Managers in Asia, Europe, and North America
- 20% Asset Price Reduction due to Climate Impact
- 40% Annual Returns Reduction due to Climate Impact
- Tech-enabled, data-driven, adoption-ready customers 10%
- Ambient's Market Share of 10%

$26B

Serviceable Available Market (SAM)

Customers ready for adoption within next 3-5 years

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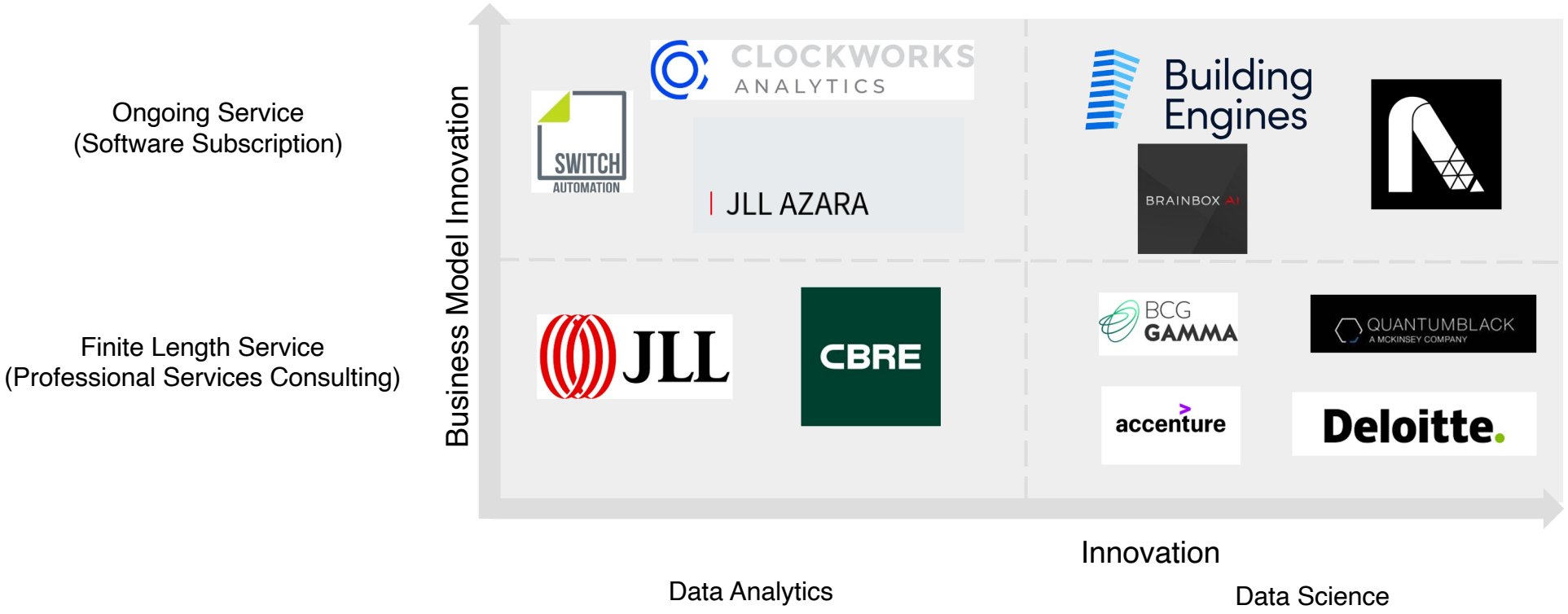
$260B

Total Addressable Market (TAM)

Global Portfolio Value and Return Losses

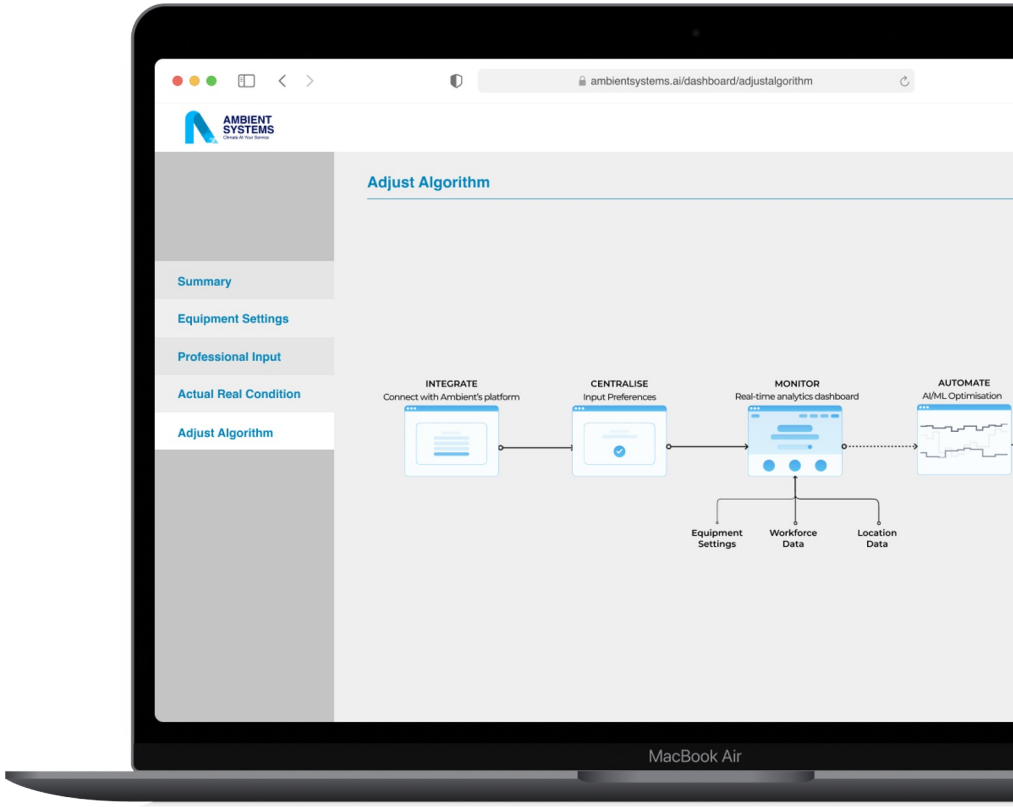
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Ambient aims to be a leader in data science solutions for decarbonization of CRE assets



Decarbonization Intelligence Platform for Commercial Real Estate

Intuitive and effective insights delivered by Ambient's **state of the art platform** and **proprietary data science methodology**



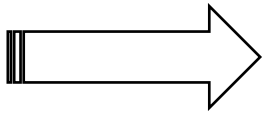
Ambient identifies the mapping between key independent and dependent variables

Independent Variables

Employment Growth in Related Sectors	Conventional
Employment Rate	Conventional
Economic GDP Growth	Conventional
New construction supply	Conventional
Inflation	Conventional
Facility Management Costs	Conventional
Building Construction Type	Conventional
Property Location	Conventional
Supporting Infrastructure	Innovative
Building Facilities Technology	Innovative
Climate Change Physical Risk	Climate
Climate Change Transition Risk	Climate

Baseline

Ambient



Dependent Variables

1. Asset Valuation & Appraisal Confidence
2. Occupancy / Vacancy
3. Operating Expense and Operating Expense Ratio

Supporting asset managers across the property lifecycle



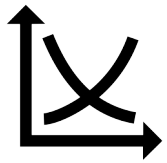
15% to 25%

Enhanced Valuation Accuracy
(to reflect climate change risk)



30%

Reduced Operating Expenses
(through operational efficiency insights)



3X

Modeling Time Savings
(compared to current process)

APPLICATIONS

Ambient supports a broad a range of CRE assets

			
Industrial	**Retail**	**Offices**	**Transportation & Logistics**
Manufacturing R&D Life Sciences and BioTech	Malls Shopping Centers Grocery Stores	High Rise Low / Mid Rise Data Centers	Public Transit Airports Warehouses and DCs

Balanced CRE optimization & decarbonization through three unique features



Flexible On-demand Subscription

No more onboarding, training and manpower reliance. Fast up-time and results through Ambient's subscription service



Simple and Affordable Pricing Plans

No more reliance on expensive consultants or in-house teams. One platform powered for all your data science use cases



Specialized AI Model Customization

Reach your business goals across energy efficiency, environment, and compliance with Ambient's state of the art AI models

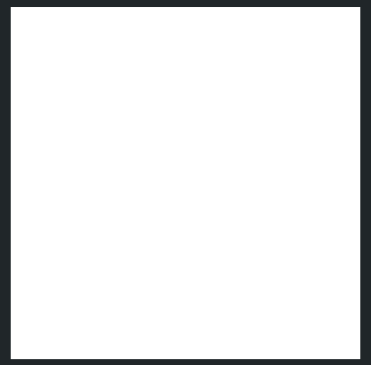
COMPANY VISION

Empower CRE leaders to achieve net zero portfolio growth through data science innovation

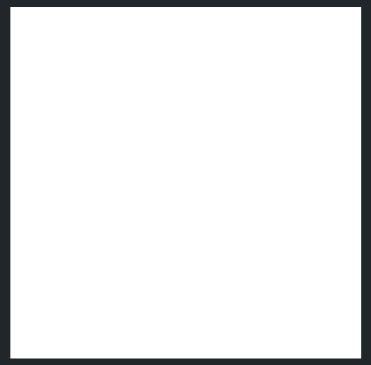


ABOUT US

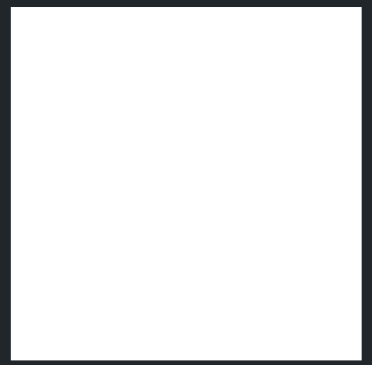
Our Team



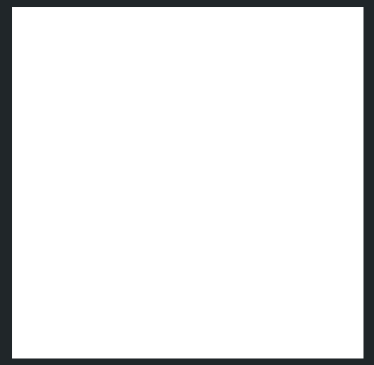
CEO
Deep tech entrepreneur with 3+ years experience in [redacted] and clean tech



UX/UI Designer
Service & UIUX Designer with background in VC & Industrial Design

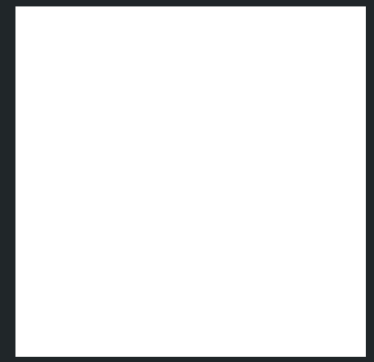


Data Scientist
[redacted] engineer with experience building cross-industry AI applications at Meta and AWS



[redacted] Developer
Experienced developer of gaming and advanced web-based applications

Mentors & Advisors



Dr. [redacted]
Experienced technology executive, investor, and educator with 35 years of leadership experience at Oracle and HP

THE ASK

Raising USD 300K Pre-Seed Round with USD 75K committed

$37.5K
RAISED

$37.5K
COMMITTED

$225K
FUNDING GAP

Use of Funds

- Runway: ~mid 2023
- Hire core team: Engineering, Design, and Marketing
- Enable international co-innovation projects
- Markets: Singapore
- WIPO and filing costs

Extra Slides

Appendix & FAQ

Hard to Find Technical Expertise for Managing Net Zero Smart Buildings

Edge Devices

Automation

User Behavior

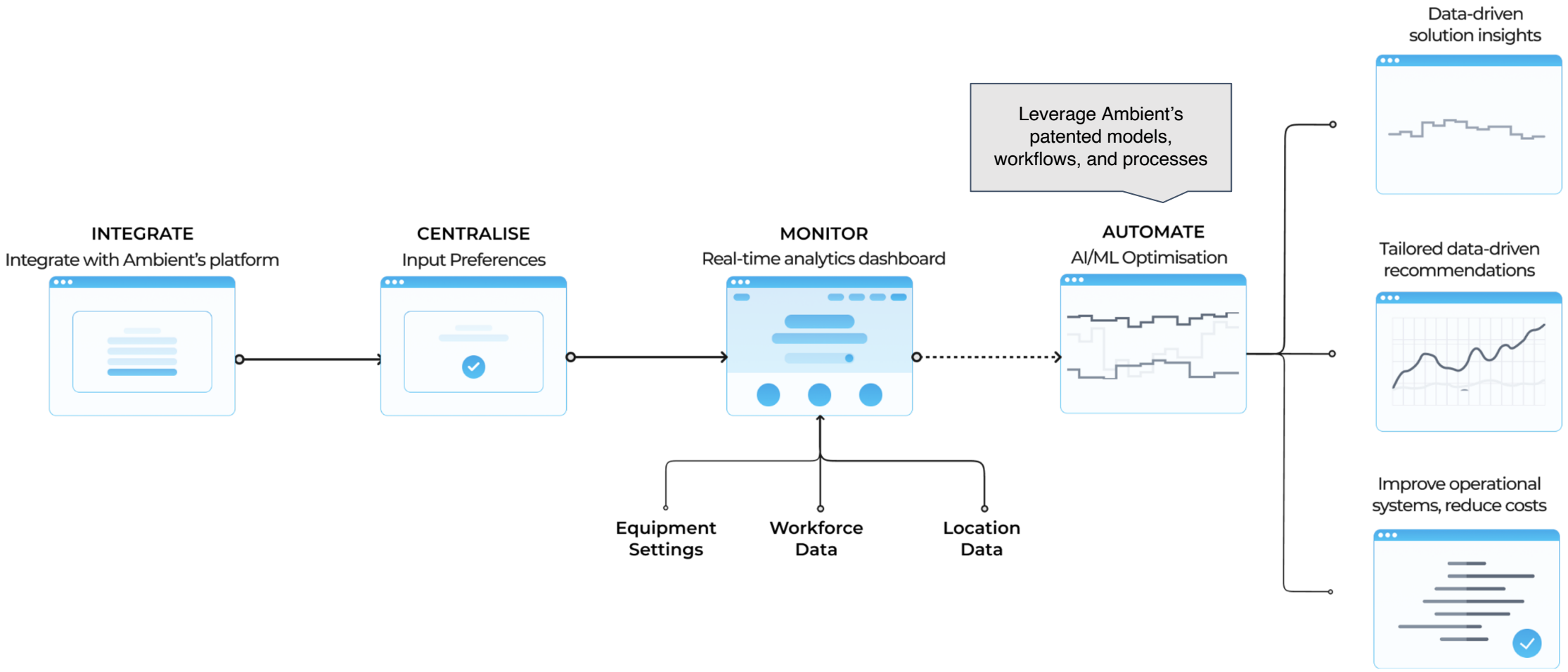


Inefficient energy consumption

Continuous operational cost

High

Data Science pipeline for Real Asset Optimization & Decarbonization



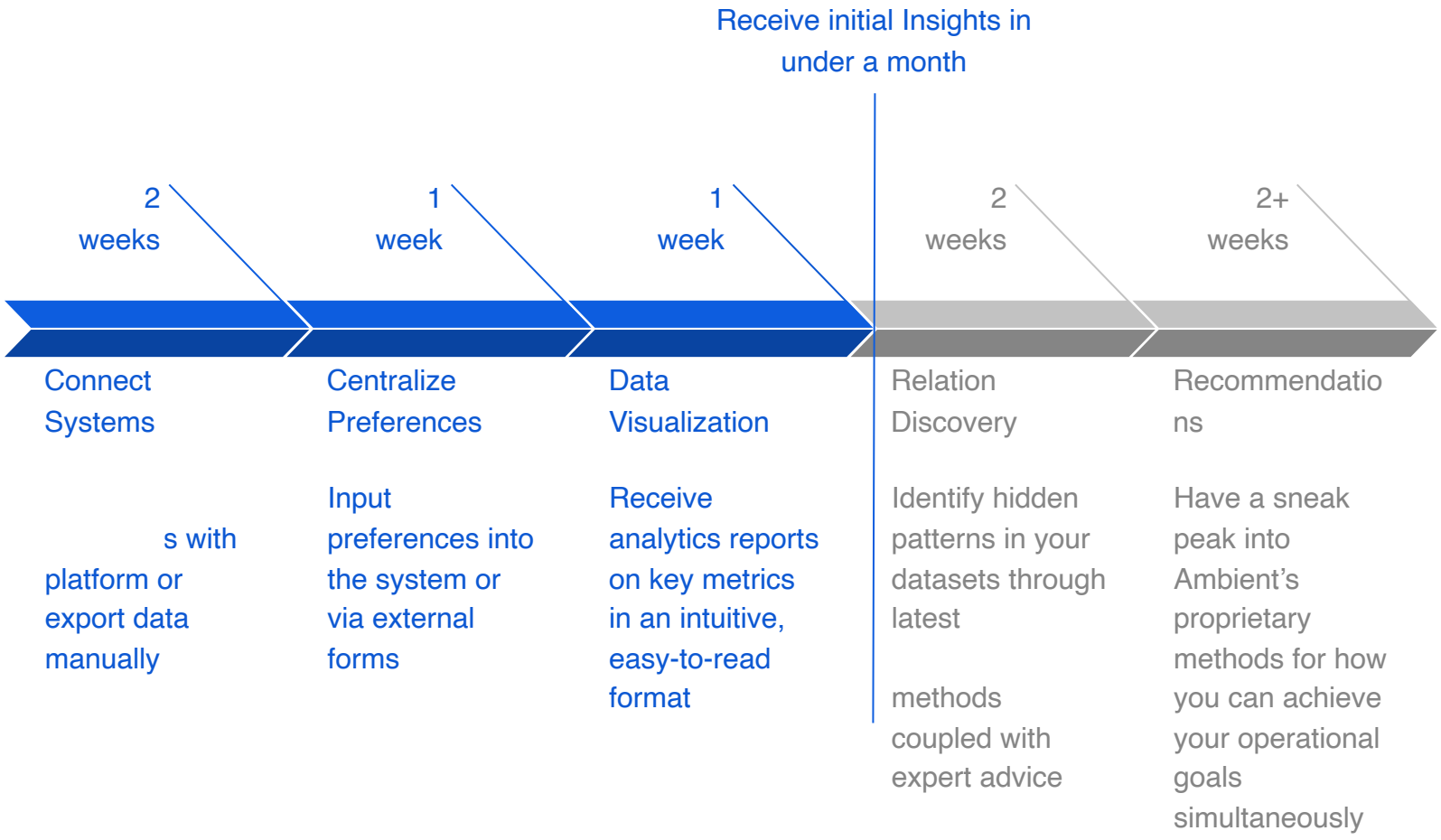
Best in class in service, cost, and customization

	Ambient Systems Data Science Subscription	Data Science Professional Services	Data Science In-House Teams
Service	On-demand Subscription	Limited Time Project	Hard to recruit Requires on-boarding Training needed Management overhead High turnover risk
Cost	Affordable	Expensive	Expensive
Customization	Specialized	Generic	Requires multiple experts to tune Technical uncertainty and risk Expensive and complex IT vendor management process Cybersecurity risks

Empowering Leading Real Estate Asset Managers and Operators in Asia and Globally

Firm	HQ	AUM in Asia
GLP	Singapore	$65B
IGIS Asset Management	Seoul, Korea	$36B
ESR	Hong Kong	$26B
Blackstone	New York, USA	$26B
Brookfield Asset Management	Toronto, Canada	$26B

POC duration can range 3 to 4 months with initial insights within 2-4 weeks from setting up the system



Tailored CRE data and insights platform, balancing productivity with sustainability

Insights



Portfolio

- Portfolio Performance
- Leasing
- Transactions



Facilities & Operations

- Decarbonization
- Operational Performance
- Maintenance & Repairs
- Energy Efficiency



Space & Workplace

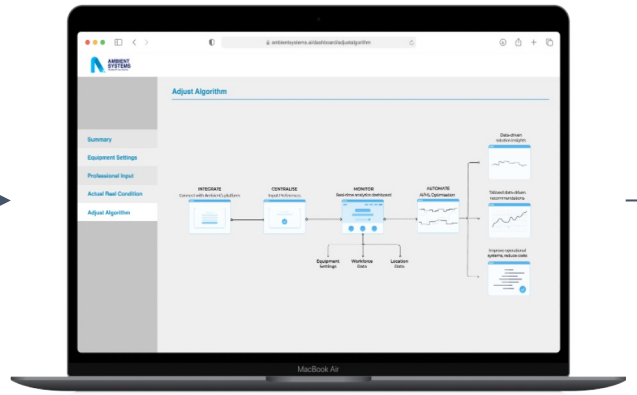
- Workplace Performance
- Occupancy
- Utilization

Platform

CRE Data Model	API Connections	Data Engineering	Data Visualization	Notifications

Not a BMS or BAS but Compatible with your and Protocols

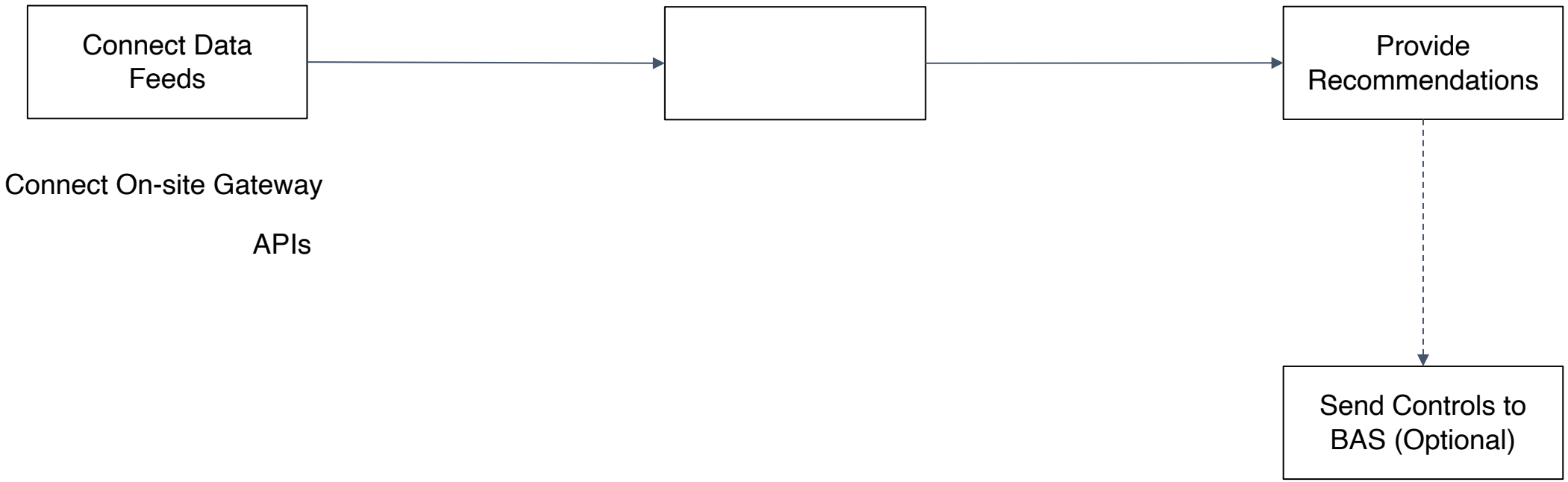
Ambient helps your facility managers make sense of BMS data and streamline



BMS Database

Siemens Building Technologies, , Honeywell, Bosch, ABB, Schneider Electric

Part 1 - Platform: end-to-end CRE property optimization and decarbonization

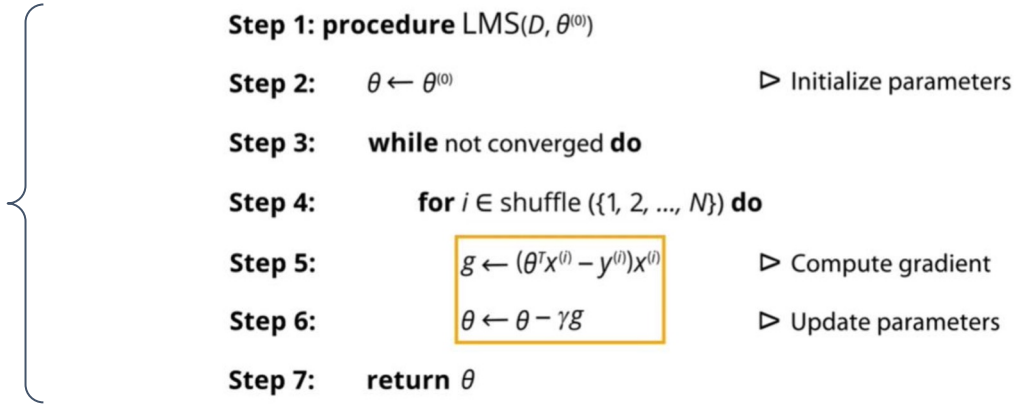


Part 2 - Models: latest models tailored for your occupiers

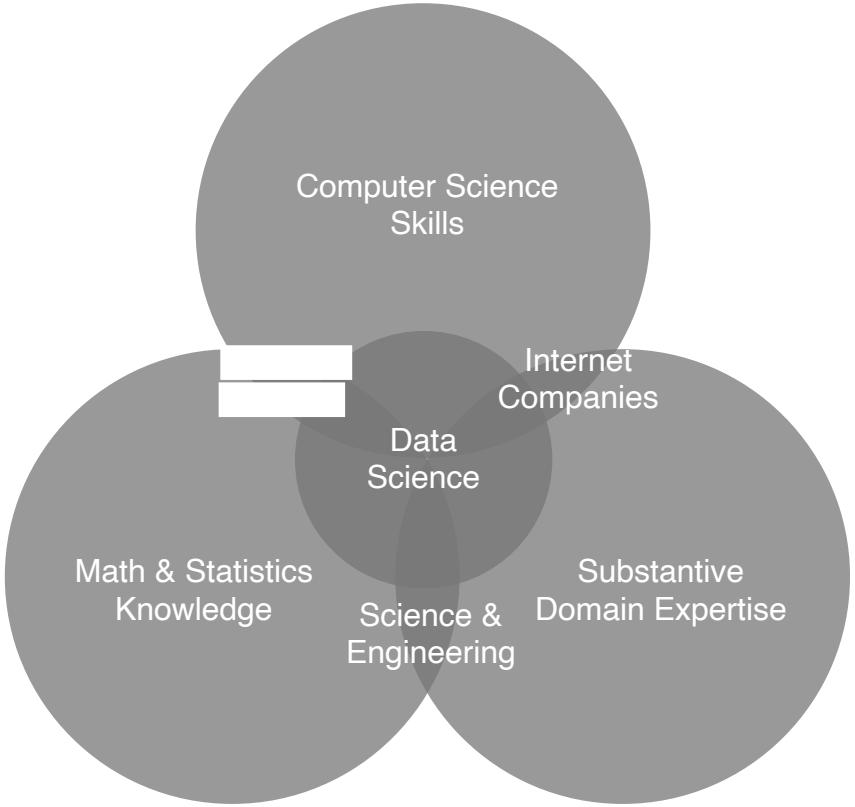
Key Steps:

1. Identify variable correlations (e.g., Energy Consumption and Temperature)
2. Formulate objective function (e.g., minimize energy consumption)
3. Formulate constraints functions (e.g., IAQ, CO2, Relative Humidity, Thermal Comfort)
4. Optional: Assign objective vector weights
5. Perform multi-objective optimization leveraging (i.e., vector optimization)

Stochastic Gradient Descent



Q: Why are Data Scientists hard to find?



Niche Requirements
Shortage of data scientists who can optimize multiple intelligent systems

Competitive Landscape
Only around 2k data scientists in Singapore and around 50k in US, with majority working for tech giants

Data Science Teams can require over 10 roles, making them too costly for most organizations

Q: What can Data Scientists do?

General Activities

- Find internal and external data to identify new opportunities or risk areas
- Develop an understanding of user needs
- Communicate findings to stakeholders
- Implement data-driven insights
- Keep a pulse on the latest technical and industry developments and how they can enable smarter business processes



Specific to Real Estate

1. Control energy efficiency
2. Balance energy savings with environmental considerations (healthy environment (IAQ), CO2, etc.)
3. Create an environment for driving productivity and revenues
4. Streamline compliance (regulations, certifications, and standards)

Q: What are Cyber Physical Systems?

Cyber Physical Systems (CPS) Components

Internet

+

Electronic Devices (IoT)

+

Physical Surroundings & Equipment

+

Users

Cyber Physical Systems (CPS) Applications

Commercial Buildings

Industrial Buildings

Transport Infrastructure

Healthcare Facilities

Smart Homes

Utilities (water, electricity, natural gas)

Q: What are Ambient's Data Science Capabilities?

Data Engineering **(Input)**	**(Use Cases)**	Data Visualization **(Output)**
• Data Selection • Data Preprocessing • Data E hment • Feature Engineering • Data Transformation • Exploratory Data Analysis	• Predictive Maintenance • Demand Forecasting • Image and Video Analytics • Inventory Optimization • Recommender Systems • **Graph Mining (Discovering insights in real world networks)*** • **Equipment Calibration*** • **Metrics Balance and Optimization***	• Moving average • Box plots • Histograms • Contours and Density Contours • Hexbin • Q-Q Plot • Parallel Sets • Treemap and hierarchy graphs • Maps and map tiles

***Only offered by Ambient**

Technologies we Use



Q: Do you offer hardware, sensors, or BMS?

We don’t sell them directly but we can help you evaluate vendors to make sure they meet your business needs and you can access the data you need.

Q: Will you recommend AC System upgrades?

Yes, we will provide recommendations where we see room for improvement.

As with the BMS, we do not sell it ourselves.

Q: How long to see the results?

The accuracy and insights will improve over time.

But you can expect initial insights within two weeks after connecting the relevant data feeds.

Q: Is my data secure?

Yes.

We work with leading vendors, such as AWS and C3.AI, to power our platform and models

We also follow national and international regulations on data protection and data privacy.